AERIS Expert Review Packet

Anomaly

Source / metric	tceq / no2
Observed / expected	19.5 / 4.01724
Time	2026-06-01 13:00 UTC
Location	29.7589, -95.0793
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	humidity	percent	7 / 336	43.99 to 100; mean 77.8158	88.84 at 2026-06-01 12:55Z; dt 5 min
asos	temperature	degC	7 / 336	23.3333 to 34; mean 28.1214	26.7778 at 2026-06-01 12:55Z; dt 5 min
asos	wind_direction	degree	7 / 336	0 to 360; mean 95.4464	240 at 2026-06-01 12:55Z; dt 5 min
asos	wind_speed	m/s	7 / 336	0 to 8.2311; mean 2.2078	1.54333 at 2026-06-01 12:55Z; dt 5 min
noaa_gfs	gh_500	m	9 / 108	5877.66 to 5901.91; mean 5890.44	5887.34 at 2026-06-01 12:00Z; dt 60 min
noaa_gfs	pbl_height	m	9 / 108	14.9006 to 2020.13; mean 669.369	72.0103 at 2026-06-01 12:00Z; dt 60 min
noaa_gfs	precipitable_water	mm	9 / 108	39.1994 to 52.7754; mean 47.5051	48.0252 at 2026-06-01 12:00Z; dt 60 min
noaa_gfs	surface_pressure	hPa	9 / 108	1006.34 to 1014.42; mean 1010.94	1012.74 at 2026-06-01 12:00Z; dt 60 min
noaa_gfs	t_850	degC	9 / 108	16.5256 to 20.4157; mean 18.2908	18.0026 at 2026-06-01 12:00Z; dt 60 min
noaa_gfs	u_10m	m/s	9 / 108	-3.67282 to 0.197017; mean -1.4755	-0.0534277 at 2026-06-01 12:00Z; dt 60 min
noaa_gfs	v_10m	m/s	9 / 108	-1.54819 to 3.78564; mean 1.4302	0.524851 at 2026-06-01 12:00Z; dt 60 min
openaq	ozone	ppm	11 / 474	-0.002 to 0.094; mean 0.0197	0.008 at 2026-06-01 13:00Z; dt 0 min
openaq	pm10	ug/m3	2 / 88	12 to 110; mean 30.2273	29 at 2026-06-01 13:00Z; dt 0 min
openaq	pm25	ug/m3	26 / 1978	-3.3 to 34.5; mean 6.0505	10.4 at 2026-06-01 13:00Z; dt 0 min
openweather	cloud_cover	percent	4 / 288	0 to 100; mean 41.5069	0 at 2026-06-01 12:57Z; dt 2.4 min
openweather	humidity	percent	4 / 288	57 to 95; mean 78.6632	89 at 2026-06-01 12:57Z; dt 2.4 min
openweather	precipitation	mm/h	4 / 288	0 to 4.86; mean 0.087	0 at 2026-06-01 12:57Z; dt 2.4 min
openweather	pressure	hPa	4 / 288	1011 to 1015; mean 1013.12	1014 at 2026-06-01 12:57Z; dt 2.4 min
openweather	temperature	degC	4 / 288	23.99 to 34.34; mean 28.4931	26.17 at 2026-06-01 12:57Z; dt 2.4 min
openweather	wind_direction	degree	4 / 288	0 to 360; mean 123.528	0 at 2026-06-01 12:57Z; dt 2.4 min
openweather	wind_speed	m/s	4 / 288	0 to 7.2; mean 2.6092	0 at 2026-06-01 12:57Z; dt 2.4 min
purpleair	pm25	ug/m3	55 / 2733	0 to 58; mean 6.0915	5.8 at 2026-06-01 13:00Z; dt 0 min

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
sentinel5p	s5p_aer_ai_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-06-01 19:42Z; dt 402.2 min
sentinel5p	s5p_ch4_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-06-01 19:42Z; dt 402.2 min
sentinel5p	s5p_co_column	mol/m^2	4 / 4	0.028772 to 0.0302015; mean 0.0295	0.0300364 at 2026-06-01 19:42Z; dt 402.2 min
sentinel5p	s5p_co_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-06-01 19:42Z; dt 402.2 min
sentinel5p	s5p_hcho_column	mol/m^2	5 / 5	0.000117197 to 0.000169173; mean 0.0001	0.000166437 at 2026-06-01 19:42Z; dt 402.2 min
sentinel5p	s5p_hcho_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-06-01 19:42Z; dt 402.2 min
sentinel5p	s5p_no2_column	mol/m^2	6 / 6	1.483e-05 to 4.38402e-05; mean 0	3.77616e-05 at 2026-06-01 19:42Z; dt 402.2 min
sentinel5p	s5p_no2_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-06-01 19:42Z; dt 402.2 min
sentinel5p	s5p_o3_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-06-01 19:42Z; dt 402.2 min
sentinel5p	s5p_so2_column	mol/m^2	5 / 5	-9.16616e-06 to 0.000116468; mean 0	1.74952e-05 at 2026-06-01 19:42Z; dt 402.2 min
sentinel5p	s5p_so2_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-06-01 19:42Z; dt 402.2 min
tceq	co	ppm	3 / 200	0.1 to 0.9; mean 0.221	0.1 at 2026-06-01 13:00Z; dt 0 min
tceq	no2	ppb	12 / 764	-2.6 to 34.4; mean 5.1717	19.5 at 2026-06-01 13:00Z; dt 0 min
tceq	so2	ppb	3 / 197	-0.5 to 1.6; mean -0.0711	0.1 at 2026-06-01 13:00Z; dt 0 min

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	humidity	06-01 00Z 82.71, 06Z 94.08, 12Z 69.79, 18Z 61.03, 06-02 00Z 82.02, 06Z 92.86, 12Z 75.71, 18Z 65.3, 06-03 00Z 80.79
asos	temperature	06-01 00Z 26.84, 06Z 24.78, 12Z 29.45, 18Z 31.54, 06-02 00Z 27.57, 06Z 25.49, 12Z 28.74, 18Z 30.41, 06-03 00Z 27.4
asos	wind_direction	06-01 00Z 134.5, 06Z 27.56, 12Z 80.98, 18Z 136.4, 06-02 00Z 137.8, 06Z 47.18, 12Z 90.48, 18Z 109, 06-03 00Z 72.86
asos	wind_speed	06-01 00Z 2.548, 06Z 0.4015, 12Z 1.343, 18Z 3.969, 06-02 00Z 2.161, 06Z 0.5408, 12Z 2.107, 18Z 4.14, 06-03 00Z 3.822
noaa_gfs	gh_500	05-31 06Z 5891, 12Z 5881, 18Z 5896, 06-01 00Z 5891, 06Z 5900, 12Z 5886, 18Z 5901, 06-02 00Z 5888, 06Z 5898, 12Z 5881, 18Z 5893, 06-03 00Z 5879
noaa_gfs	pbl_height	05-31 06Z 301.4, 12Z 168, 18Z 1575, 06-01 00Z 841, 06Z 270.2, 12Z 87.32, 18Z 1602, 06-02 00Z 687.1, 06Z 106.8, 12Z 53.16, 18Z 1552, 06-03 00Z 788.3
noaa_gfs	precipitable_water	05-31 06Z 40.73, 12Z 43.7, 18Z 46.08, 06-01 00Z 47.28, 06Z 46.83, 12Z 46.55, 18Z 49.73, 06-02 00Z 51.2, 06Z 50.41, 12Z 50.79, 18Z 49.42, 06-03 00Z 47.35
noaa_gfs	surface_pressure	05-31 06Z 1009, 12Z 1011, 18Z 1011, 06-01 00Z 1009, 06Z 1012, 12Z 1011, 18Z 1011, 06-02 00Z 1010, 06Z 1012, 12Z 1012, 18Z 1012, 06-03 00Z 1011
noaa_gfs	t_850	05-31 06Z 20.06, 12Z 18.84, 18Z 16.84, 06-01 00Z 18.62, 06Z 18.58, 12Z 17.89, 18Z 17.44, 06-02 00Z 18.89, 06Z 18.52, 12Z 17.57, 18Z 16.96, 06-03 00Z 19.28
noaa_gfs	u_10m	05-31 06Z -1.024, 12Z -0.636, 18Z -1.552, 06-01 00Z -2.957, 06Z -0.8569, 12Z -0.4756, 18Z -1.812, 06-02 00Z -2.422, 06Z -0.47, 12Z -0.6052, 18Z -2.115, 06-03 00Z -2.779

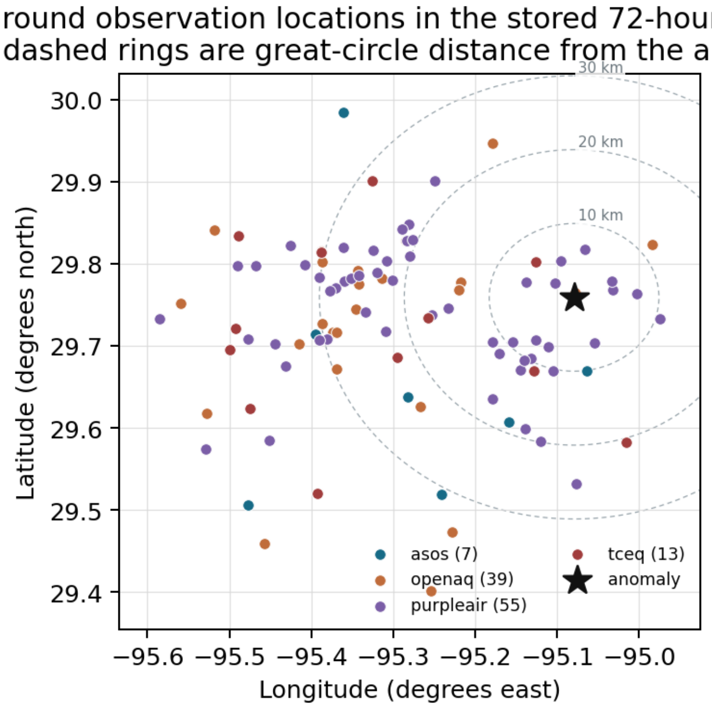
Source	Metric	Values
noaa_gfs	v_10m	05-31 06Z 2.709, 12Z 1.596, 18Z 2.178, 06-01 00Z 2.889, 06Z 1.982, 12Z 0.8204, 18Z 0.7199, 06-02 00Z 2.836, 06Z 1.55, 12Z -0.9006, 18Z -1.094, 06-03 00Z 1.877
openaq	ozone	06-01 06Z 0.009439, 12Z 0.01643, 18Z 0.04763, 06-02 00Z 0.01729, 06Z 0.007485, 12Z 0.01031, 18Z 0.03205, 06-03 00Z 0.01141
openaq	pm10	06-01 06Z 31.42, 12Z 44.58, 18Z 30.83, 06-02 00Z 25.17, 06Z 22.08, 12Z 31.17, 18Z 28.42, 06-03 00Z 24
openaq	pm25	06-01 06Z 6.639, 12Z 6.042, 18Z 6.91, 06-02 00Z 5.455, 06Z 5.123, 12Z 6.98, 18Z 5.333, 06-03 00Z 6.503
openweather	cloud_cover	05-31 00Z 28.75, 06Z 30.62, 12Z 57.67, 18Z 49.58, 06-01 00Z 15.08, 06Z 40.5, 12Z 45.88, 18Z 43.75, 06-02 00Z 24.38, 06Z 25.62, 12Z 55.62, 18Z 84.38, 06-03 00Z 6.25
openweather	humidity	05-31 00Z 79.85, 06Z 89.83, 12Z 74.67, 18Z 64.71, 06-01 00Z 81.21, 06Z 92.42, 12Z 74.92, 18Z 65.08, 06-02 00Z 80.08, 06Z 91.29, 12Z 79, 18Z 70.17, 06-03 00Z 84.25
openweather	precipitation	05-31 00Z 0, 06Z 0, 12Z 0, 18Z 0, 06-01 00Z 0, 06Z 0.00625, 12Z 0, 18Z 0.009167, 06-02 00Z 0, 06Z 0, 12Z 0, 18Z 0.8808, 06-03 00Z 0.89
openweather	pressure	05-31 00Z 1012, 06Z 1012, 12Z 1013, 18Z 1012, 06-01 00Z 1013, 06Z 1014, 12Z 1014, 18Z 1012, 06-02 00Z 1013, 06Z 1014, 12Z 1015, 18Z 1013, 06-03 00Z 1014
openweather	temperature	05-31 00Z 26.84, 06Z 24.85, 12Z 29.24, 18Z 31.97, 06-01 00Z 27.41, 06Z 24.98, 12Z 29.75, 18Z 32.42, 06-02 00Z 28.19, 06Z 25.75, 12Z 29.16, 18Z 31.33, 06-03 00Z 26.95
openweather	wind_direction	05-31 00Z 166.5, 06Z 120.8, 12Z 154.7, 18Z 137.1, 06-01 00Z 148, 06Z 70.21, 12Z 106.4, 18Z 122.7, 06-02 00Z 168.4, 06Z 57.21, 12Z 120.6, 18Z 125.5, 06-03 00Z 72.25
openweather	wind_speed	05-31 00Z 3.601, 06Z 1.95, 12Z 2.965, 18Z 4.544, 06-01 00Z 3.666, 06Z 0.9775, 12Z 1.11, 18Z 3.975, 06-02 00Z 3.167, 06Z 0.9167, 12Z 1.902, 18Z 2.681, 06-03 00Z 2.728
purpleair	pm25	06-01 00Z 5.966, 06Z 7.794, 12Z 5.71, 18Z 6.781, 06-02 00Z 5.263, 06Z 5.66, 12Z 6.695, 18Z 4.382, 06-03 00Z 7.605
sentinel5p	s5p_aer_ai_granule_available	05-31 18Z 1, 06-01 18Z 1, 06-02 18Z 1
sentinel5p	s5p_ch4_granule_available	05-31 18Z 1, 06-01 18Z 1, 06-02 18Z 1
sentinel5p	s5p_co_column	06-01 18Z 0.03012, 06-02 18Z 0.02882
sentinel5p	s5p_co_granule_available	05-31 18Z 1, 06-01 18Z 1, 06-02 18Z 1
sentinel5p	s5p_hcho_column	05-31 18Z 0.0001186, 06-01 18Z 0.0001645, 06-02 18Z 0.0001692
sentinel5p	s5p_hcho_granule_available	05-31 18Z 1, 06-01 18Z 1, 06-02 18Z 1
sentinel5p	s5p_no2_column	05-31 18Z 1.667e-05, 06-01 18Z 4.08e-05, 06-02 18Z 4.07e-05
sentinel5p	s5p_no2_granule_available	05-31 18Z 1, 06-01 18Z 1, 06-02 18Z 1
sentinel5p	s5p_o3_granule_available	05-31 18Z 1, 06-01 18Z 1, 06-02 18Z 1
sentinel5p	s5p_so2_column	05-31 18Z 0.0001165, 06-01 18Z 1.316e-05, 06-02 18Z -8.155e-06
sentinel5p	s5p_so2_granule_available	05-31 18Z 1, 06-01 18Z 1, 06-02 18Z 1
tceq	co	05-31 06Z 0.1833, 12Z 0.2167, 18Z 0.22, 06-01 00Z 0.25, 06Z 0.2833, 12Z 0.2529, 18Z 0.2278, 06-02 00Z 0.2, 06Z 0.2333, 12Z 0.2056, 18Z 0.1667, 06-03 00Z 0.2
tceq	no2	05-31 06Z 3.488, 12Z 3.185, 18Z 2.902, 06-01 00Z 3.521, 06Z 5.504, 12Z 6.443, 18Z 5.943, 06-02 00Z 4.576, 06Z 5.149, 12Z 7.831, 18Z 6.415, 06-03 00Z 9.535
tceq	so2	05-31 06Z -0.2188, 12Z -0.2467, 18Z -0.1556, 06-01 00Z -0.1556, 06Z -0.1778, 12Z 0.1889, 18Z 0.08333, 06-02 00Z -0.1444, 06Z -0.1294, 12Z -0.01176, 18Z 0.1556, 06-03 00Z -0.1167

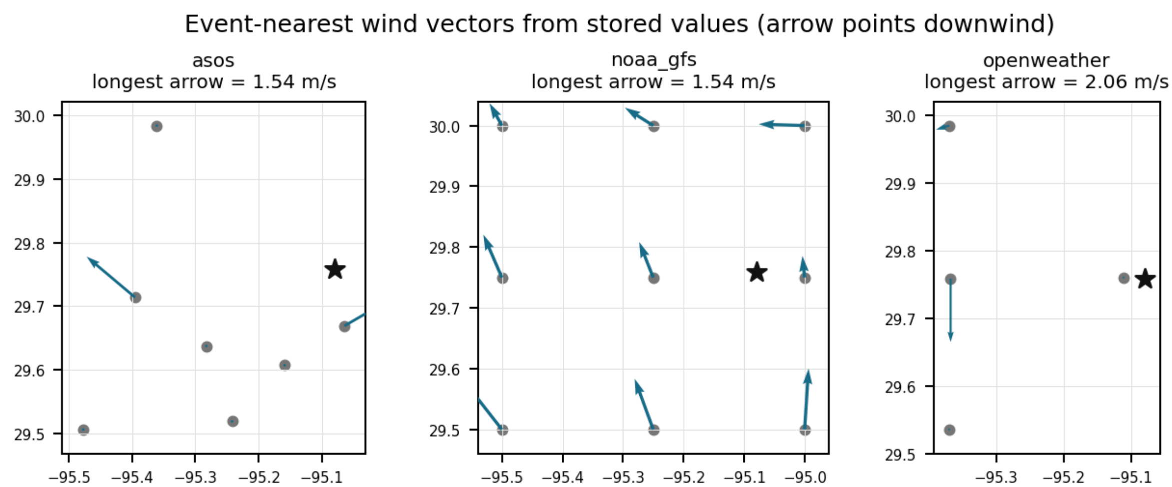
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

Source	Metric	Values
asos	humidity	T41 (10.087 km) 80.61, EFD (18.523 km) 78.37, HOU (23.813 km) 76.53, MCJ (30.885 km) 76.61, LVJ (30.963 km) 77.57, IAH (36.939 km) 79.26, AXH (47.612 km) 75.72
asos	temperature	T41 (10.087 km) 27.8, EFD (18.523 km) 28.22, HOU (23.813 km) 28.44, MCJ (30.885 km) 28.52, LVJ (30.963 km) 28, IAH (36.939 km) 28, AXH (47.612 km) 27.89
asos	wind_direction	T41 (10.087 km) 110.4, EFD (18.523 km) 93.56, HOU (23.813 km) 114.1, MCJ (30.885 km) 110.4, LVJ (30.963 km) 73.88, IAH (36.939 km) 99.8, AXH (47.612 km) 66.73
asos	wind_speed	T41 (10.087 km) 2.352, EFD (18.523 km) 2.275, HOU (23.813 km) 2.446, MCJ (30.885 km) 3.355, LVJ (30.963 km) 1.764, IAH (36.939 km) 2.089, AXH (47.612 km) 1.249
noaa_gfs	gh_500	gfs:29.75,-95.00 (7.724 km) 5891, gfs:29.75,-95.25 (16.505 km) 5891, gfs:30.00,-95.00 (27.874 km) 5890, gfs:29.50,-95.00 (29.797 km) 5891, gfs:30.00,-95.25 (31.451 km) 5889, gfs:29.50,-95.25 (33.184 km) 5891, gfs:29.75,-95.50 (40.62 km) 5890, gfs:30.00,-95.50 (48.614 km) 5889, +1 more
noaa_gfs	pbl_height	gfs:29.75,-95.00 (7.724 km) 560.7, gfs:29.75,-95.25 (16.505 km) 790.3, gfs:30.00,-95.00 (27.874 km) 489.8, gfs:29.50,-95.00 (29.797 km) 563.8, gfs:30.00,-95.25 (31.451 km) 767.4, gfs:29.50,-95.25 (33.184 km) 557.9, gfs:29.75,-95.50 (40.62 km) 821.2, gfs:30.00,-95.50 (48.614 km) 899.5, +1 more
noaa_gfs	precipitable_water	gfs:29.75,-95.00 (7.724 km) 48.29, gfs:29.75,-95.25 (16.505 km) 47.47, gfs:30.00,-95.00 (27.874 km) 47.95, gfs:29.50,-95.00 (29.797 km) 48.24, gfs:30.00,-95.25 (31.451 km) 47.06, gfs:29.50,-95.25 (33.184 km) 47.92, gfs:29.75,-95.50 (40.62 km) 46.89, gfs:30.00,-95.50 (48.614 km) 46.38, +1 more
noaa_gfs	surface_pressure	gfs:29.75,-95.00 (7.724 km) 1012, gfs:29.75,-95.25 (16.505 km) 1011, gfs:30.00,-95.00 (27.874 km) 1011, gfs:29.50,-95.00 (29.797 km) 1013, gfs:30.00,-95.25 (31.451 km) 1010, gfs:29.50,-95.25 (33.184 km) 1012, gfs:29.75,-95.50 (40.62 km) 1010, gfs:30.00,-95.50 (48.614 km) 1008, +1 more
noaa_gfs	t_850	gfs:29.75,-95.00 (7.724 km) 18.33, gfs:29.75,-95.25 (16.505 km) 18.36, gfs:30.00,-95.00 (27.874 km) 18.21, gfs:29.50,-95.00 (29.797 km) 18.31, gfs:30.00,-95.25 (31.451 km) 18.35, gfs:29.50,-95.25 (33.184 km) 18.33, gfs:29.75,-95.50 (40.62 km) 18.2, gfs:30.00,-95.50 (48.614 km) 18.23, +1 more
noaa_gfs	u_10m	gfs:29.75,-95.00 (7.724 km) -1.485, gfs:29.75,-95.25 (16.505 km) -1.461, gfs:30.00,-95.00 (27.874 km) -1.371, gfs:29.50,-95.00 (29.797 km) -1.827, gfs:30.00,-95.25 (31.451 km) -1.592, gfs:29.50,-95.25 (33.184 km) -1.595, gfs:29.75,-95.50 (40.62 km) -1.179, gfs:30.00,-95.50 (48.614 km) -1.129, +1 more
noaa_gfs	v_10m	gfs:29.75,-95.00 (7.724 km) 1.594, gfs:29.75,-95.25 (16.505 km) 1.313, gfs:30.00,-95.00 (27.874 km) 1.511, gfs:29.50,-95.00 (29.797 km) 1.715, gfs:30.00,-95.25 (31.451 km) 1.187, gfs:29.50,-95.25 (33.184 km) 1.487, gfs:29.75,-95.50 (40.62 km) 1.53, gfs:30.00,-95.50 (48.614 km) 1.176, +1 more
openaq	ozone	2800 (0.618 km) 0.01859, 332 (10.969 km) 0.02093, 339 (11.673 km) 0.02041, 2039 (13.673 km) 0.02098, 3378 (17.424 km) 0.01691, 2046 (21.211 km) 0.01764, 325 (22.309 km) 0.01844, 320 (23.434 km) 0.01547, +3 more
openaq	pm10	271 (10.969 km) 22.16, 13380082 (17.424 km) 38.3
openaq	pm25	268 (10.969 km) 8.564, 13787195 (13.552 km) 7.521, 2040 (13.673 km) 10.82, 3379 (17.424 km) 9.373, 8967479 (17.481 km) 7.309, 2071331 (21.211 km) 12.17, 8967217 (22.812 km) 5.622, 13906493 (23.001 km) 1.143, +18 more
openweather	cloud_cover	grid:east (3.07 km) 39.33, grid:center (27.835 km) 42.31, grid:south (37.464 km) 28.82, grid:north (37.632 km) 55.57
openweather	humidity	grid:east (3.07 km) 82.14, grid:center (27.835 km) 77.75, grid:south (37.464 km) 76.74, grid:north (37.632 km) 78.03
openweather	precipitation	grid:east (3.07 km) 0.04167, grid:center (27.835 km) 0.1393, grid:south (37.464 km) 0.02014, grid:north (37.632 km) 0.1471
openweather	pressure	grid:east (3.07 km) 1013, grid:center (27.835 km) 1013, grid:south (37.464 km) 1013, grid:north (37.632 km) 1013
openweather	temperature	grid:east (3.07 km) 28.16, grid:center (27.835 km) 28.93, grid:south (37.464 km) 28.31, grid:north (37.632 km) 28.57
openweather	wind_direction	grid:east (3.07 km) 108, grid:center (27.835 km) 133.3, grid:south (37.464 km) 128, grid:north (37.632 km) 124.8
openweather	wind_speed	grid:east (3.07 km) 2.38, grid:center (27.835 km) 3.244, grid:south (37.464 km) 2.398, grid:north (37.632 km) 2.415
purpleair	pm25	148709 (2.995 km) 7.718, 213701 (4.738 km) 7.422, 268171 (4.996 km) 5.27, 166399 (5.166 km) 8.194, 186271 (6.02 km) 8.954, 186277 (6.577 km) 10.28, 222815 (6.641 km) 5.076, 222593 (7.346 km) 5.756, +47 more
tceq	co	482011039 (10.998 km) 0.1433, 482011035 (17.437 km) 0.1697, 482011052 (30.401 km) 0.3493

Source	Metric	Values
tceq	no2	482011015 (0.0 km) 6.466, 482010026 (6.597 km) 8.403, 482011039 (10.998 km) 3.982, 482011035 (17.437 km) 7.867, 482011050 (20.507 km) 1.688, 482010416 (22.313 km) 4.892, 482010024 (28.571 km) 0.2489, 482011052 (30.401 km) 9.786, +4 more
tceq	so2	482011039 (10.998 km) 0.1758, 482011035 (17.437 km) -0.06765, 482010051 (41.001 km) -0.303





Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). The labeling guide that comes with this packet has the definitions and marking rules. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Claim 1 of 35

The air quality anomaly in Houston, Texas is caused by low wind speeds.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 2 of 35

The temperature on June 1, 2026 was relatively high, with a mean value of 28.4931°C.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 3 of 35

Ground-level nitrogen dioxide concentrations at the TCEQ monitoring station in Houston spiked to 19.5 ppb at 13:00 UTC on June 1, 2026.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 4 of 35

A fresh local NOx plume from nearby combustion sources such as roadway traffic, industrial equipment, marine activity, or rail operations is a likely physical cause of the 19.5 ppb NO2 spike because the anomaly was highly localized, ozone at a nearby sensor was low at 0.008 ppm at the same hour, and co-located SO2 and nearby CO did not show comparably large increases.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 5 of 35

NOAA GFS model analyses indicated a very shallow planetary boundary layer height below 90 meters around 12:00 UTC on June 1, 2026, which restricted vertical mixing of low-level pollutants.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 6 of 35

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 7 of 35

The tceq NO2 monitor at 29.759, -95.079 recorded 19.5 ppb at 2026-06-01T13:00:00+00:00 in Houston, Texas.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 8 of 35

Ground-level nitrogen dioxide concentrations measured by TCEQ reached 19.5 ppb at 13:00 UTC on June 1, 2026, aligned with Sentinel-5P satellite observations showing elevated NO2 column densities around 4.08e-05 mol/m^2.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 9 of 35

A broader regional urban-industrial NO2 enhancement likely contributed background support for the local spike because Sentinel-5P showed elevated Houston-area tropospheric NO2 later on 2026-06-01, with an NO2 column near 3.78e-05 mol/m^2 and a 18Z mean around 4.08e-05 mol/m^2 that was much higher than the previous day.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 10 of 35

The temperature during the anomaly was around 26.17 degC, which is lower than the mean value of 28.4931 degC.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 11 of 35

Sentinel-5P satellite observations indicated an elevated average NO_2 column density of 4.08e-05 mol/m^2 over the area during the June 1, 2026 18Z window compared to 1.667e-05 mol/m^2 on May 31.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 12 of 35

The air quality anomaly in Houston, Texas is caused by high levels of NO_2.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 13 of 35

The anomaly is a result of a combination of factors including high temperatures, low humidity, and stagnant atmospheric conditions, leading to poor air mixing and increased pollutant concentrations.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 14 of 35

The reported tceq NO_2 anomaly at 2026-06-01T13:00:00+00:00 had an expected value of 4.017241379310344 ppb and a z-score of 9.00539499896256, which is labeled severe.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 15 of 35

Shallow mixing and weak winds are a likely physical cause of the NO2 buildup because the Houston area had very low morning transport and ventilation near the anomaly time, including ASOS wind speed of 1.54 m/s near 12:55 UTC and GFS planetary boundary layer height near 72 m at 12 UTC.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 16 of 35

Weak horizontal wind dispersion restricted the transport and dilution of NO2 away from the TCEQ surface monitor, leading to a localized peak in ground-level concentration.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 17 of 35

A severe thunderstorm or tornado event occurred in the area, stirring up pollutants and reducing air quality.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 18 of 35

At 2026-06-01T13:00:00+00:00 near the anomaly location, openaq ozone was 0.008 ppm at a sensor 0.618 km away, openaq PM2.5 was 10.4 ug/m3 at a sensor 10.969 km away, and purpleair PM2.5 was 5.8 ug/m3 at a sensor 2.995 km away.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 19 of 35

The air quality anomaly in Houston, Texas is characterized by elevated levels of PM2.5 and NO2.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 20 of 35

Sentinel-5P satellite observations detected elevated tropospheric nitrogen dioxide column densities over the Houston metropolitan area on June 1, 2026 relative to May 31, 2026.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 21 of 35

The anomaly occurred on June 1st at 13:00:00+00:00 (0.0 min away) and was detected by sensor 148709, located approximately 2.995 km from the anomaly.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 22 of 35

The pollutant mix near the NO2 spike is most consistent with a fresh local NOx plume because nearby ozone was low at 0.008 ppm while PM2.5 remained modest near 5.8 to 10.4 ug/m3 and nearby CO and SO2 did not show comparably large increases.

Calm-wind flag: wind direction unstable under calm conditions. noaa_gfs: event wind 0.5275634302233442 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave 0.3515518403576108 m/s; n=108 wind readings); openweather: event wind 0.0 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave -0.7021895498075934 m/s; n=288 wind readings). Cutoff floor: confirmed 2026-07-24.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 23 of 35

The anomaly is caused by a nearby industrial facility releasing excessive amounts of particulate matter into the atmosphere.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 24 of 35

Meteorological conditions during the morning of June 1, 2026 were characterized by a shallow planetary boundary layer height below 100 meters and light surface wind speeds around 1.5 m/s, suppressing vertical and horizontal dispersion.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 25 of 35

There is no evidence to suggest that the air quality anomaly in Houston, Texas was caused by any specific industrial or agricultural activities.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 26 of 35

The air quality anomaly is related to particulate matter (PM2.5) with a value of 7.718 ug/m3, which is elevated compared to the mean value of 6.0915 ug/m3.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 27 of 35

ASOS surface observations confirmed stagnant local dispersion conditions, recording average morning wind speeds as low as 0.40 m/s prior to the nitrogen dioxide peak.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 28 of 35

The local fresh-emission-plume hypothesis is supported because the anomalous TCEQ monitor measured 19.5 ppb NO2 at 13 UTC while the nearest OpenAQ ozone measurement was only 0.008 ppm at the same time, a combination consistent with recently emitted NO scavenging ozone near the source.

Calm-wind flag: wind direction unstable under calm conditions. noaa_gfs: event wind 0.5275634302233442 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave 0.3515518403576108 m/s; n=108 wind readings); openweather: event wind 0.0 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave -0.7021895498075934 m/s; n=288 wind readings). Cutoff floor: confirmed 2026-07-24.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 29 of 35

The stagnation-limited-dilution hypothesis is supported because surface winds near the anomaly were weak at about 1.54 m/s in ASOS observations near 12:55 UTC and the GFS planetary boundary layer height near 12 UTC was only about 72 m, both favoring accumulation of near-surface NO2.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 30 of 35

The Houston TCEQ monitor recorded a severe local NO2 spike of 19.5 ppb at 2026-06-01T13:00:00+00:00 versus an expected 4.02 ppb, indicating an event that greatly exceeded the monitor's usual level.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 31 of 35

A shallow planetary boundary layer height below 100 meters combined with low surface wind speeds trapped local emissions near the ground around 12:00-13:00 UTC on June 1, 2026.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 32 of 35

Houston-area meteorology at the time of the NO2 spike favored accumulation of primary emissions because surface winds were weak near the site and the modeled planetary boundary layer height was very low at about 72 m around 12 UTC.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 33 of 35

A severe nitrogen dioxide (NO2) concentration anomaly of 19.5 ppb was recorded at coordinates (29.759, -95.079) on June 1, 2026, at 13:00 UTC, exceeding the expected value of 4.02 ppb with a z-score of 9.01.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 34 of 35

Concentrated nitrogen dioxide emissions from early-morning traffic and industrial combustion accumulated rapidly under restricted boundary layer mixing.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 35 of 35

The broader urban-or-industrial enhancement hypothesis has partial support because Sentinel-5P showed elevated Houston-area NO2 columns at 18Z on 2026-06-01 relative to 2026-05-31, but the exact 13 UTC spike is not strongly supported as a combustion-rich regional plume because nearby TCEQ CO was only 0.1 ppm and nearby TCEQ SO2 was only 0.1 ppb at 13 UTC.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.